

Python Dictionary Basics (Beginner Guide)

1 What is a Dictionary?

A **dictionary** stores data as **key** → **value** pairs.

```
employee = {  
    "id": 101,  
    "name": "edwin",  
    "salary": 45000,  
    "dept": "qa"  
}
```

- Keys must be unique
 - Values can be any data type
-

2 Common Dictionary Methods

♦ **keys()** → Get all keys

```
for key in employee.keys():  
    print(key)
```

Output example:

```
id  
name  
salary  
dept
```

♦ **values()** → Get all values

```
for val in employee.values():  
    print(val)
```

♦ **items()** → Get key–value pairs

```
for key, val in employee.items():
```

```
print(key, val)
```

Output:

```
id 101
name edwin
salary 45000
dept qa
```

♦ **get(key)** → **Safe value lookup**

```
print(employee.get("email"))
```

- Returns **None** if key doesn't exist
- No error

With default value:

```
print(employee.get("email", "dummy@gmail.com"))
```

Output:

```
dummy@gmail.com
```

♦ **pop(key)** → **Remove a key**

```
employee.pop("dept")
print(employee)
```

- Removes the key and returns its value
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3 Difference: **get()** vs **[]**

✗ **Using brackets (unsafe)**

```
employee["email"] # Raises KeyError
```

✓ Using get() (safe)

`employee.get("email")` # Returns None

4 See all dictionary methods

`print(dir(dict))`

Shows all built-in methods available for dictionaries.

✓ Key Takeaways

- Dictionary = key → value storage
 - Use:
 - `keys()` → only keys
 - `values()` → only values
 - `items()` → both
 - `get()` → safe lookup
 - `pop()` → remove key
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